



Tecnológico Nacional de México

Instituto Tecnológico de Saltillo

Computo en la nube

4.3 Seguridad (hardening) en servidor web Apache 2

Docente: Miguel Salazar Del Bosque

**Alumno
Diego Josué González de León**

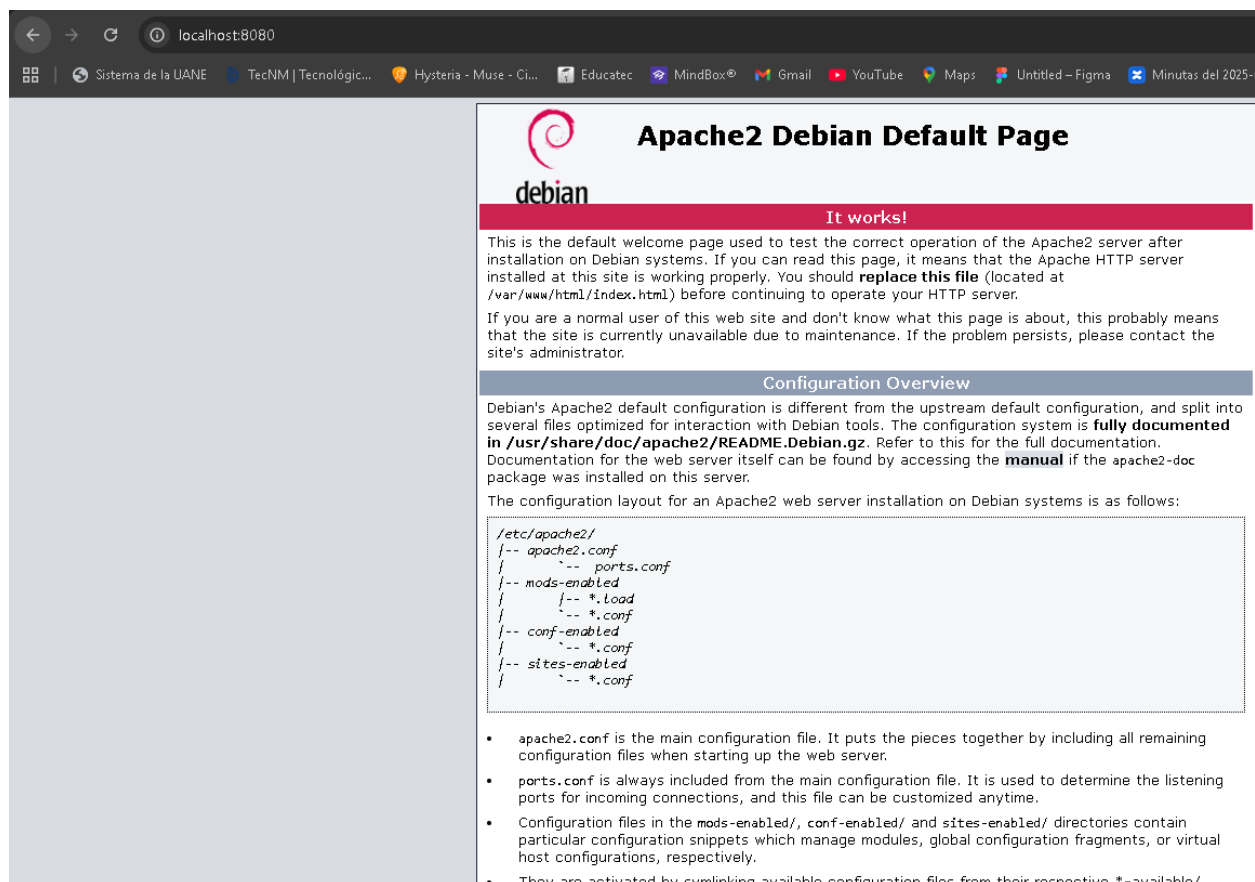
20 de marzo de 2026

Paso 1: Apache corriendo

```
diego@debian:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-03-20 09:08:27 EDT; 8min ago
  Invocation: 82854e5a284f4bf8934789ed4e5df884
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 883 (apache2)
      Tasks: 6 (limit: 4639)
    Memory: 21M (peak: 21.4M)
       CPU: 242ms
    CGroup: /system.slice/apache2.service
            └─883 /usr/sbin/apache2 -k start
              └─885 /usr/sbin/apache2 -k start
                └─886 /usr/sbin/apache2 -k start
                  └─887 /usr/sbin/apache2 -k start
                    └─888 /usr/sbin/apache2 -k start
                      └─889 /usr/sbin/apache2 -k start

Mar 20 09:08:27 debian systemd[1]: Starting apache2.service - The Apache HTTP Server...
Mar 20 09:08:27 debian apachectl[821]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive dynamically to determine the server's name. See the documentation for the 'ServerName' directive for details.
Mar 20 09:08:27 debian systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-20/20 (END)
```

Paso 2: Verificar en el navegador



localhost:8080

Apache2 Debian Default Page

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Debian systems. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Debian's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Debian tools. The configuration system is **fully documented** in `/usr/share/doc/apache2/README.Debian.gz`. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Debian systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective `*-available/`

Paso 3: Ocultar informacion sensible

```
ComputoNube APS clon [Corriendo] - Oracle VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

#
# ServerTokens
# This directive configures what you return as the Server HTTP response
# Header. The default is 'Full' which sends information about the OS-Type
# and compiled in modules.
# Set to one of: Full | OS | Minimal | Minor | Major | Prod
# where Full conveys the most information, and Prod the least.
#ServerTokens Minimal
ServerTokens Prod
#ServerTokens Full

#
# Optionally add a line containing the server version and virtual host
# name to server-generated pages (internal error documents, FTP directory
# listings, mod_status and mod_info output etc., but not CGI generated
# documents or custom error documents).
# Set to "EMail" to also include a mailto: link to the ServerAdmin.
# Set to one of: On | Off | EMail
#ServerSignature Off
ServerSignature Off_

#
# Allow TRACE method
#
# Set to "extended" to also reflect the request body (only for testing and
# diagnostic purposes).
#
# Set to one of: On | Off | extended
TraceEnable Off
#TraceEnable On
```

```
diego@debianz:~$ sudo a2enconf security
Conf security already enabled
diego@debianz:~$ sudo systemctl restart apache2
diego@debianz:~$
```

Paso 4: Deshabilitar modulos inecesarios

```
diego@debianz:~$ sudo apache2ctl -M
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Loaded Modules:
  core_module (static)
  so_module (static)
  watchdog_module (static)
  http_module (static)
  log_config_module (static)
  logio_module (static)
  version_module (static)
  unixd_module (static)
  access_compat_module (shared)
  alias_module (shared)
  auth_basic_module (shared)
  authn_core_module (shared)
  authn_file_module (shared)
  authz_core_module (shared)
  authz_host_module (shared)
  authz_user_module (shared)
  autoindex_module (shared)
  deflate_module (shared)
  dir_module (shared)
  env_module (shared)
  filter_module (shared)
  mime_module (shared)
  mpm_prefork_module (shared)
  negotiation_module (shared)
  php_module (shared)
  reqtimeout_module (shared)
  setenvif_module (shared)
  status_module (shared)
diego@debianz:~$
```

```
diego@debianz:~$ sudo a2dismod autoindex status proxy proxy_http
Module autoindex already disabled
Module status already disabled
Module proxy already disabled
Module proxy_http already disabled
diego@debianz:~$
```

Paso 5: Configurar HTTPS


```
ComputoNube APS clon [Corriendo] - Oracle VirtualBox
Archivo  Máquina  Ver  Entrada  Dispositivos  Ayuda

DocumentRoot /var/www/html

# Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
# error, crit, alert, emerg.
# It is also possible to configure the loglevel for particular
# modules, e.g.
#LogLevel info ssl:warn

ErrorLog ${APACHE_LOG_DIR}/error.log
CustomLog ${APACHE_LOG_DIR}/access.log combined

# For most configuration files from conf-available/, which are
# enabled or disabled at a global level, it is possible to
# include a line for only one particular virtual host. For example the
# following line enables the CGI configuration for this host only
# after it has been globally disabled with "a2disconf".
#Include conf-available/serve-cgi-bin.conf

# SSL Engine Switch:
# Enable/Disable SSL for this virtual host.
SSLEngine on

# A self-signed (snakeoil) certificate can be created by installing
# the ssl-cert package. See
# /usr/share/doc/apache2/README.Debian.gz for more info.
# If both key and certificate are stored in the same file, only the
# SSLCertificateFile directive is needed.
SSLCertificateFile /etc/apache2/ssl/apache.crt
SSLCertificateKeyFile /etc/apache2/ssl/apache.key_

# Server Certificate Chain:
# Point SSLCertificateChainFile at a file containing the
# concatenation of PEM encoded CA certificates which form the
# certificate chain for the server certificate. Alternatively
# the referenced file can be the same as SSLCertificateFile
# when the CA certificates are directly appended to the server
# certificate for convinience.
#SSLCertificateChainFile /etc/apache2/ssl.crt/server-ca.crt

# ... (rest of the file content) ...
```

```
diego@debianz:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled
diego@debianz:~$ sudo systemctl restart apache2
diego@debianz:~$ S_
```

No segurohttps://192.168.56.10

Sistema de la UANE

TecNM | Tecnológic...

>>

Todos los favor



La conexión no es privada

Es posible que un atacante esté intentando robarte la información de **192.168.56.10** (por ejemplo, contraseñas, mensajes o tarjetas de crédito). [Obtén más información sobre esta advertencia](#)

NET::ERR_CERT_AUTHORITY_INVALID



Activa la [protección mejorada](#) para obtener el nivel más alto de seguridad de Chrome

Ocultar detalles avanzados

Volver a un sitio seguro

Este servidor no pudo probar que su dominio es **192.168.56.10**; el sistema operativo de la computadora no confía en el certificado de seguridad. Es posible que esto se deba a una configuración incorrecta o a que un atacante interceptó la conexión.

[Continuar a 192.168.56.10 \(no seguro\)](#)

```
diego@debianz:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled
diego@debianz:~$ sudo systemctl restart apache2
diego@debianz:~$ sudo ss -tlnl

Netid      State      Recv-Q     Send-Q      Local Address:Port
udp        UNCONN     0           0           10.0.2.15:68
udp        UNCONN     0           0           [fd17:625c:f037:2:a00:27ff:fe2d:7ebc]:546
udp        UNCONN     0           0           [fd17:625c:f037:2:70b9:ce8a:8355:46d7]:546
udp        UNCONN     0           0           [fe80::d8f3:2a48:725f:a4fa]:%enp0s3:546
tcp        LISTEN     0           80          127.0.0.1:3306
tcp        LISTEN     0           128         0.0.0.0:22
tcp        LISTEN     0           511         *:443
tcp        LISTEN     0           128         [::]:22
tcp        LISTEN     0           511         *:80
diego@debianz:~$ sudo ufw allow 80
Rule added
Rule added (v6)
diego@debianz:~$ sudo ufw allow 443
Rule added
Rule added (v6)
diego@debianz:~$ sudo ufw enable
Firewall is active and enabled on system startup
diego@debianz:~$
```


Paso 7: Logs y Monitoreo

```
diego@debianz:~$ sudo tail -f /var/log/apache2/access.log
10.0.2.2 - - [20/Mar/2026:10:11:40 -0400] "\x16\x03\x01\x06\x02\x01" 400 432 "-" "-"
10.0.2.2 - - [20/Mar/2026:10:11:41 -0400] "\x16\x03\x01\x06\x02\x01" 400 432 "-" "-"
10.0.2.2 - - [20/Mar/2026:10:11:41 -0400] "\x16\x03\x01\x07\x12\x01" 400 432 "-" "-"
10.0.2.2 - - [20/Mar/2026:10:11:41 -0400] "\x16\x03\x01\x06\x02\x01" 400 432 "-" "-"
10.0.2.2 - - [20/Mar/2026:10:11:41 -0400] "\x16\x03\x01\x07\x12\x01" 400 432 "-" "-"
192.168.56.1 - - [20/Mar/2026:10:13:02 -0400] "-" 408 0 "-" "-"
192.168.56.1 - - [20/Mar/2026:10:13:17 -0400] "GET / HTTP/1.1" 200 6069 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64)
rome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:10:13:17 -0400] "GET /icons/openlogo-75.png HTTP/1.1" 200 6046 "https://192.168.56.10/"
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:10:13:17 -0400] "GET /favicon.ico HTTP/1.1" 404 458 "https://192.168.56.10/" "Mozilla/5.0
t/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
192.168.56.1 - - [20/Mar/2026:10:14:08 -0400] "-" 408 0 "-" "-"
^C
diego@debianz:~$ sudo tail -f /var/log/apache2/error.log
[Fri Mar 20 09:47:47.805088 2026] [mpm_prefork:notice] [pid 2051:tid 2051] AH00170: caught SIGWINCH, shutting down graceful
[Fri Mar 20 09:47:47.920376 2026] [mpm_prefork:notice] [pid 2267:tid 2267] AH00163: Apache/2.4.66 (Debian) configured
[Fri Mar 20 09:47:47.920434 2026] [core:notice] [pid 2267:tid 2267] AH00094: Command line: '/usr/sbin/apache2'
[Fri Mar 20 10:09:13.518259 2026] [mpm_prefork:notice] [pid 2267:tid 2267] AH00170: caught SIGWINCH, shutting down graceful
[Fri Mar 20 10:09:13.612713 2026] [ssl:warn] [pid 2602:tid 2602] AH01906: 127.0.1.1:443:0 server certificate is a CA certificate
[Fri Mar 20 10:09:13.612763 2026] [ssl:warn] [pid 2602:tid 2602] AH01909: 127.0.1.1:443:0 server certificate does NOT
[Fri Mar 20 10:09:13.640833 2026] [ssl:warn] [pid 2612:tid 2612] AH01906: 127.0.1.1:443:0 server certificate is a CA certificate
[Fri Mar 20 10:09:13.640857 2026] [ssl:warn] [pid 2612:tid 2612] AH01909: 127.0.1.1:443:0 server certificate does NOT
[Fri Mar 20 10:09:13.644940 2026] [mpm_prefork:notice] [pid 2612:tid 2612] AH00163: Apache/2.4.66 (Debian) OpenSSL/3.5
[Fri Mar 20 10:09:13.644980 2026] [core:notice] [pid 2612:tid 2612] AH00094: Command line: '/usr/sbin/apache2'
^C
diego@debianz:~$
```